

| Question | Answer                                                                                                                                        | Marks       |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 6(a)(i)  | lamp needs 'high' power/'large' current/'large' voltage                                                                                       | <b>B1</b>   |
|          | op-amp can deliver only a small current/small voltage                                                                                         | <b>B1</b>   |
| 6(a)(ii) | correct symbol for relay coil connected between output and earth                                                                              | <b>B1</b>   |
|          | switch between mains supply and lamp                                                                                                          | <b>B1</b>   |
| 6(b)(i)  | vary light intensity at which lamp is switched on/off                                                                                         | <b>B1</b>   |
| 6(b)(ii) | so that relay operates for only one current/voltage direction<br><b>or</b><br>so that relay/lamp operates for either dark or light conditions | <b>B1</b>   |
| 6(c)     | when light level increases, LDR resistance decreases                                                                                          | <b>B1</b>   |
|          | ( $R_{\text{LDR}}$ low,) so $V^- > V^+$ , so $V_{\text{OUT}}$ negative/−5 V (must be consistent with B1 mark)                                 | <b>M1</b>   |
|          | <b>or</b>                                                                                                                                     |             |
|          | when light level decreases, LDR resistance increases                                                                                          | <b>(B1)</b> |
|          | ( $R_{\text{LDR}}$ high,) so $V^- < V^+$ , so $V_{\text{OUT}}$ is positive/+5 V (must be consistent with B1 mark)                             | <b>(M1)</b> |
|          | lamp comes on as light level decreases<br><b>or</b><br>lamp goes off as light level increases                                                 | <b>A1</b>   |

9702/41

## Cambridge International AS/A Level – Mark Scheme

May/June 2017

**PUBLISHED**

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|----------|------------------------------------------------------------------|-----------|
| 7(c)     | (magnetic) force (always) normal to velocity/direction of motion | <b>M1</b> |